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Electronic TONE GENERATOR SYSTEM

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It would be cool to build an electronic tone generator system yourself. Described here is a small tone generator based on pulse-width modulation (PWM) concept in which a piezo buzzer generates multiple tones with the help of Arduino software programming.

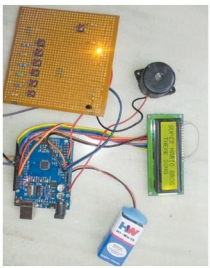

Fig. 1: Author's prototype

You can see how the device works, including the menu of preset musical tones, in the author's prototype shown in Fig. 1. There are five musical tones already programmed in this project. Technically, tones four and five in this project are the same, which are left to readers to create better ones.

Writing a musical code was not the goal of the project. Basic coding for main loops and individual tone loops is easy to adapt to your own ideas, so feel free to change the code as per your requirement.

Circuit and working

Circuit diagram of the electronic tone generator is shown in Fig. 2. It is based on Arduino Uno board (BOARD1), liquid crystal display (LCD1), transistor BC547 (T1), a buzzer (PZ1) and a few other components.

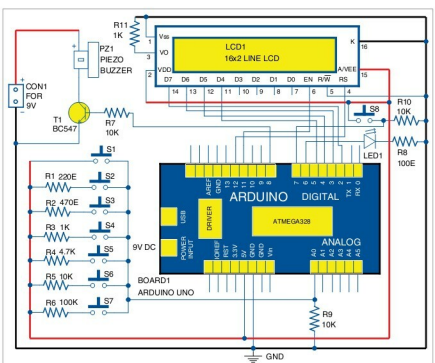

Fig. 2: Circuit diagram of the tone generator system

Tone key switches (S1 through S7) are connected using a resistor ladder. The resistors (R1 through R6) are placed in a sequential order, connecting each switch to 5V power supply. Resistor values (in ohms) used are 100k, 10k, 4.7k, 1k, 470-ohm and 220-ohm.

S1 is directly connected to 5V power supply. One terminal of switch S2 is connected to 5V supply through R1 and the other terminal to pin A0 of Board1. Switches S3 through S7 are also connected to analogue pin A0 of Arduino Uno board in a similar manner.

The Arduino program (tone_generator.ino) defines the musical notes and frequencies associated with each tone key. Change the frequency values or add additional switches to fully customise your project.

LCD screen. Connect the LCD1 pins according to the circuit diagram shown in Fig. 2. Resistor R11 connected at pin 3 of LCD1 is used as the contrast control of the screen. Pin 15 of LCD1 is connected to 5V power supply and pin 16 is connected to ground for backlight on the LCD screen. LCD1 is configured in 4-bit mode. Its pins D4, D5, D6 and D7, register select (RS) and enable (EN) are connected to pins 5, 4, 3, 2, 12 and 11, respectively, of the Arduino Uno R3 board. Read/write (R/$\overline{W}$) pin is connected to ground.

Connect one terminal of menu switch S8 to 5V and the other terminal to ground through R10. Also, connect it to pin 7 of the Arduino.

Glowing of LED1 indicates that you are in Menu mode and the tone keys have been disabled. LED1 turns on once S8 is pressed. Connect the anode (positive) of LED1 to pin 6 of Arduino Uno and cathode to ground through R8.

To integrate the piezo buzzer (PZ1) through transistor T1, connect the base of the transistor to Arduino's pin 8 through R7. Connect piezo